

Miracle Chimamanda Ndubuisi

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[Email](#) | [LinkedIn](#) | [GitHub](#) | [Website](#) | [Google Scholar](#)

PROFILE

Security Analyst with experience across detection engineering, incident response, digital forensics, application security, cloud security, and DevSecOps within the fintech sector.

Discovered more than 40 production security vulnerabilities across mobile applications, web platforms, APIs, payment systems, and cloud infrastructure, and led the forensic investigation of a targeted social-engineering intrusion. My research interests have developed from these operational security challenges, with a focus on detection engineering, cloud security, and empirical security measurement. I am seeking a thesis-based research master's to investigate how security controls and detection capabilities can be measured, evaluated, and improved in real-world environments.

RESEARCH INTERESTS

- Detection Engineering and Security Analytics
- Cloud and Infrastructure Security
- Empirical Security Measurement
- Network and Systems Security
- Identity and Access Security
- Digital Forensics and Incident Response

EDUCATION

Alex Ekwueme Federal University, Ebonyi State, Nigeria

Bachelor of Science (Education) in Physics, First Class Honours | Sep 2017 - Dec 2021

Undergraduate Thesis: *Awareness of Artificial Intelligence among Pre-service Teachers in Ebonyi State*

- Designed and conducted a quantitative study involving 200+ participants, including survey instrument development, data collection, and statistical analysis using SPSS.
- Research contributed to a peer-reviewed publication on pre-service teachers' awareness, willingness, and readiness to adopt artificial intelligence.

PUBLICATIONS

Manuscript Under Review

Ndubuisi, M. C., Abanum, C., & Adigwu, D. U. *AI: A Danger to an Illiterate Society*. Manuscript under review at *Knowledge and Information Systems*.

Peer-Reviewed Publications

Chiago, C. A., Ndubuisi, M. C., et al. (2024). Evaluating the Impact of AI on Medical Physics: A Mini

Review. *REJOST*, 4(5), 22–43.

Agbo, J. N., Nmadu, J., Ndubuisi, M. C., et al. (2022). Awareness, Willingness, and Readiness of Pre-Service Teachers to Deploy AI. *AJSTME*, 8(1), 25–31.

PROFESSIONAL EXPERIENCE

Security Analyst

Hizo Technology Limited, Asaba, Nigeria

May 2026 - Present

Lead an independent security function spanning detection engineering, incident response, security assessment, access governance, and cloud/infrastructure security within a fintech environment.

- Detection engineering: Deployed and hardened a production Wazuh Security Information and Event Management (SIEM) platform with centralized log collection, file-integrity monitoring, access controls, and MITRE ATT&CK mapping; developed and tuned detection capabilities for operational monitoring and threat hunting.
- Incident response and digital forensics: Led the investigation of a targeted social-engineering intrusion, reconstructing the attack from initial access through fraudulent payout and producing technical forensic and executive reports for management and legal review.
- Security assessment: Conducted security reviews across mobile applications, web platforms, APIs, payment systems, and cloud infrastructure using OWASP and MITRE ATT&CK methodologies, with findings documented through reproducible evidence, severity assessment, and remediation guidance.
- Identity and access security: Designed a role-based access-control model covering 15 organisational roles and 12 application modules, alongside privileged-access controls across security and infrastructure platforms.
- Security architecture and governance: Developed security policies, threat models, supply-chain security controls, and secure infrastructure practices covering credential management, CI/CD, dependency risk, monitoring, and incident response.

Independent Security Analyst (Contract)

Vent Africa, Remote, Nigeria

May 2026 – Present

Provide independent security assessments and incident response support across web applications, infrastructure, cloud services, and access controls.

- Investigate security incidents and infrastructure compromises using server logs, system artefacts, network evidence, and application data to reconstruct activity and identify security gaps.
- Conduct penetration testing and security assessments across web applications, APIs, cloud infrastructure, and supporting services, documenting vulnerabilities with reproducible evidence, severity assessments, and remediation recommendations.
- Review identity, access, and privileged-account configurations to identify excessive permissions, exposed resources, and weaknesses in administrative controls.
- Support security architecture and infrastructure hardening decisions, including monitoring, exposure reduction, access control, and incident response improvements.

Research and Cybersecurity Analyst & Junior Mobile Application Developer

Hizo Technology Limited, Asaba, Nigeria

Jan 2026 – May 2026

Built the organisation's security research and mobile-testing capability while contributing to the Flutter application codebase.

- Conducted a seven-layer security assessment of a Flutter fintech application using OWASP MASTG methodology, identifying 40+ vulnerabilities, including a complete account-takeover chain spanning local storage, transport security, anti-tampering controls, and server-side session management.
- Conducted grey-box security testing of a multi-currency payment API across staging and production environments, identifying a critical transaction-artefact replay vulnerability capable of enabling unauthorised wallet crediting.
- Developed a pre-implementation threat model for a multi-currency payment feature, identifying 28 risks across payment integration, foreign-exchange processing, multi-country Know Your Customer (KYC) processes, cross-border transaction flows, and authentication before production deployment.
- Investigated organisational exposure to CVE-2026-48019 across internal and third-party Laravel applications, assessing applicability and potential impact.
- Contributed to the Flutter application codebase, applying secure-development practices informed by findings from application security assessments.

DevSecOps Engineer Intern

Hizo Technology Limited , Asaba, Nigeria

Jul 2025 – Dec 2025

Supported cloud infrastructure, deployment, monitoring, and security operations across staging and production environments, providing the infrastructure foundation for my subsequent transition into cybersecurity.

- Hardened staging and production servers through secure configuration, access controls, TLS configuration, and Cloudflare-based infrastructure protections.
- Provisioned DigitalOcean infrastructure using Terraform, including networking, security controls, and compute-resource configuration.
- Containerised applications using Docker and supported CI/CD automation through GitHub Actions across staging and production environments.
- Implemented infrastructure monitoring with Datadog to improve system observability and operational visibility.

SELECTED SECURITY & RESEARCH PROJECTS

Multi-Source SIEM & Detection Engineering Lab: Wazuh | [Github](#)

Designed and built an isolated multi-source security monitoring environment to investigate log collection, detection behaviour, and security telemetry across heterogeneous systems.

- Deployed Wazuh with Linux and Windows endpoints and an OPNsense firewall, centralising endpoint and network telemetry within a single detection environment.
- Integrated Sysmon and Linux authentication telemetry and configured OPNsense syslog forwarding to support process, authentication, and network-level visibility.
- Developed and validated detection scenarios using controlled activity, including failed authentication attempts, discovery commands, suspicious process execution, and network events.

- Used the environment to examine practical detection-engineering challenges including telemetry coverage, rule behaviour, alert quality, and mapping observed activity to MITRE ATT&CK techniques.

TECHNICAL EXPERTISE

Security Engineering: Vulnerability Assessment & Penetration Testing, Detection Engineering, Threat Modelling, Incident Response, Digital Forensics, Security Monitoring, Identity & Access Management

Detection & Security Monitoring: Wazuh, Sysmon, auditd, SIEM, MITRE ATT&CK, Log Analysis, File Integrity Monitoring, Network Traffic Analysis

Application & Mobile Security: Burp Suite, Mobile Security Framework (MobSF), Frida, Open Worldwide Application Security Project (OWASP) Web Security Testing Guide, OWASP Mobile Application Security Verification Standard (MASVS), OWASP Mobile Application Security Testing Guide (MASTG), API Security Testing

Cloud & Infrastructure: Amazon Web Services (AWS), DigitalOcean, Terraform, Docker, Kubernetes, Cloudflare, Linux, Nginx

DevSecOps & Observability: GitHub Actions, ArgoCD, Datadog, Prometheus, Grafana, Node Exporter

Networking & Systems: TCP/IP, DNS, TLS, HTTP/HTTPS, tcpdump, Wireshark, OPNsense/pfSense

Programming & Scripting: Python, Bash, Structured Query Language (SQL), Dart/Flutter

Research & Analysis: Quantitative Research, Survey Design, Statistical Analysis (SPSS), Technical Research, Security Documentation

ADDITIONAL ACADEMIC & TEACHING EXPERIENCE

Laboratory Assistant (Part-time)

Alex Ekwueme Federal University, Nigeria | Nov 2020 – Mar 2021

Supported undergraduate physics laboratory activities, including experimental setup, data collection, and guidance of students during laboratory-based projects.

Physics & Mathematics Teacher | 2022 – 2023

Taught senior secondary Physics and Mathematics, developed instructional materials, and prepared students for WAEC and NECO examinations.

CERTIFICATIONS & PROFESSIONAL TRAINING

- **CompTIA Security+** | In progress; exam scheduled October 2026
- **Build a Secure Google Cloud Network** | Google Cloud Skill Badge, 2026
- **Implement Cloud Security Fundamentals** | Google Cloud Skill Badge, 2026
- **Cloud Security & DevOps** | Digital Witch Academy, 2025
- **Cloud Computing** | 3 Million Technical Talent (3MTT) Programme, Federal Ministry of Communications, Innovation & Digital Economy, Nigeria, 2024
- **IBM Cybersecurity Analyst Professional Certificate** | IBM via Coursera, 2023

LANGUAGES

English | Fluent (IELTS 7.0, CEFR C1)

Igbo | Native proficiency

German | Basic proficiency (A1)